GoLoc Technology Stack

Every technology in the GoLoc platform, where it runs and what it does. Status as of 30 October 2026.

Overview

Layer	Main technology	Lives in
Backend API	Python 3.13, Django 5.2, Django REST Framework	backend/
Data and jobs	PostgreSQL 16, Redis 7, Celery	Server (Docker)
Mobile apps	React Native, Expo SDK 58, TypeScript	traveler-mob/, captain-mob/, mobile-core/
Web consoles	React, TypeScript, Vite	admin-desk/, agency-desk/
Website	HTML, CSS on Cloudflare	landing/ (goloc.in)
AI	Groq (gpt-oss-120b), Anthropic Claude (claude-opus-5-5)	backend/apps/common/llm.py
Deployment	Docker Compose, nginx, Cloudflare, EAS	docker-compose.yml, deploy/

1. Backend: the server (backend/)

Technology	What it does
Python 3.13	Programming language of the backend
Django 5.2	Main framework: users, captains, agencies, bookings, payments and business rules
Django REST Framework	The API (/api/v1/...) used by all apps and consoles
SimpleJWT	Login tokens; each token carries a role: traveler, captain, admin or agency
django-cors-headers	Allows the web consoles to call the API
python-decouple	Reads settings and secrets from the .env file
Gunicorn	Production web server that runs Django
WhiteNoise	Serves static files (CSS/JS) for Django admin
Pillow	Image processing for photo uploads
requests	Calls outside services: Twilio, Razorpay, Wikimedia

Backend modules (backend/apps/): accounts, captains, agencies, bookings, trips, itineraries, cities, discover, payments, chat, notifications, social (reels), reviews, support (SOS), adminops, assist (AI), common.

2. Database, cache and background jobs

Technology	Where	What it does
SQLite	Developer PC	Simple file database for local testing
PostgreSQL 16	Server	Production database for all data
Redis 7	Server	Cache (fast captain search) and job queue
Celery worker	Server	Background tasks: notifications, AI copy, wallet credits
Celery beat	Server	Scheduled jobs: expire unanswered requests, inactivity holds

3. Mobile apps

Technology	What it does
React Native + Expo SDK 58	One codebase builds both Android and iOS apps
TypeScript	Typed JavaScript that catches errors early
Expo Router	Screens and navigation
TanStack React Query	Fetches and caches data from the API
Zustand	App state, such as the signed-in user
Axios	HTTP client that talks to the API
expo-secure-store	Stores login tokens safely on the phone
expo-notifications	Push notifications through Expo's push service
expo-location	Live trip location and SOS
expo-image, image-picker, document-picker	Photos and KYC document uploads
expo-video, video-thumbnails	Reels
react-native-webview	Shows maps inside the app
Fraunces + Manrope	Brand fonts (Google Fonts)
mobile-core (@goloc/core)	Code shared by both apps: UI components, API client, calendar, map
npm workspaces	Links the shared package into both apps

Two apps: traveler-mob builds **GoLoc** (in.goloc.traveler); captain-mob builds **GoLoc Captain** (in.goloc.captain). Both talk to the same backend.

4. Web consoles

App	Technology	Used by
admin-desk	React, TypeScript, Vite, Axios	GoLoc team: approvals, payouts, SOS, Verified badge, agencies
agency-desk	React, TypeScript, Vite, Axios	Agencies: captains, trips, cars, bookings, wallet
Django admin	Built into Django	Raw database access for emergencies
traveler-desk, captain-desk	React	Older web versions, currently frozen

5. Website (landing/)

Technology	What it does
HTML + CSS	goloc.in pages: home, partner form, terms, privacy, refunds, contact, captain agreement
Cloudflare (Workers/Pages, wrangler)	Hosts goloc.in, DNS and free SSL

6. Artificial intelligence

Technology	Role
Groq, model openai/gpt-oss-120b	Default AI provider: fast and low cost
Anthropic Claude, model claude-opus-5-5	Alternative provider, selected with AI_PROVIDER=anthropic
AI gateway (apps/common/llm.py)	Every AI feature goes through one gateway, so the provider can be switched with one setting

AI feature	Where	What it does
Itinerary writer	itineraries/ai.py	Drafts the day-by-day trip plan and poster for captains. Falls back to a template without AI.
Edit with AI	assist/	Improves any text field (trip descriptions, bios) with tone and length presets
Discover feed copy	discover/	Writes titles and taglines for seasonal trips on the traveler home, monthly in the background. Ranking (season and distance) is not AI.

If no AI key is set, every feature still works without AI.

7. Outside services

Service	Used for	Status
Twilio Verify	OTP login by SMS	Code ready; needs account keys
Razorpay	Payments and refunds	Code ready in test mode; needs live keys
Expo Push	Push notifications	Ready; works in EAS builds
OpenStreetMap + Leaflet	Maps (free, no Google Maps)	Working
Wikimedia Commons	Real place photos for the Discover feed	Working
Cloudflare R2 / AWS S3 (django-storages, boto3)	Photo, reel and KYC file storage in production	Code ready; needs a bucket
Insurance	₹49 per guest per day fee	Not connected; set the fee to ₹0 until a partner is signed

8. Deployment and tools

Technology	What it does
Docker + Docker Compose	Runs the API, worker, scheduler, PostgreSQL, Redis and nginx on the server
nginx	Front door: routes /api to Django and serves the admin and agency consoles and media
Hostinger VPS (Ubuntu 24.04, India)	Planned launch server; move to AWS later
Cloudflare	DNS, HTTPS (Origin Certificate) and protection for api.goloc.in
EAS (Expo Application Services)	Builds the mobile apps for Play Store and App Store
Git + GitHub	Source code (Malik478b9/GoLoc)
GitHub Actions	Planned CI/CD: tests, then automatic server deploy
Django test runner	97 automated backend tests